Linear Algebra—Winter/Spring 2026

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Problem Set 6

Due: 8 PM, Tuesday March 10, 2026

Instructions

a. *Timeline.* In general, expect the first "half" of a problem set to be released by Tuesday nights (or whenever the first lecture of the week takes place). Expect the full problem set to be released by Friday nights (or whenever the second lecture of the week takes place). The full assignment will, *typically*, be due on Tuesday nights, 8 PM.

b. *Punctuality.* Aim to get the assignment done three days prior to the deadline. We will not be able to entertain any requests for extensions. If you seek clarification from us and we do not respond on time, do the following: (a) assume whatever seems most reasonable, and solve the question under that assumption and (b) ensure you have evidence for making the clarification request and report the incident to one of the PhD TAs.

c. *Resources.* Feel free to use any resource you like (including generative AI tools) to get help but please make sure you understand what you finally write and submit. The TAs may ask you to explain the reasoning behind your response if something appears suspicious.

d. *Graded vs Practice Questions.* Problems numbered using Arabic numerals (i.e. $1, 2, 3, \ldots$) constitute assignment problems that will be graded. Problems numbered using Roman numerals (i.e. i,ii,iii,...) are practice problems that will not be graded.

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1. Let V and W be vector spaces over the field F and let T be a linear transformation from V to W. Suppose the vectors $\{\alpha_1 \ldots \alpha_k\}$ form a basis for the null space of T. Let $\{\alpha_{k+1}, \ldots \alpha_n\}$ be vectors such that $\{\alpha_1, \ldots, \alpha_n\}$ are a basis for V. Show that $\{T\alpha_{k+1}, \ldots T\alpha_n\}$ are linearly independent and span the range of T.

2. Suppose V and W are finite-dimensional vector spaces. Prove the following.

a. If $\mathsf{dim}\ V > \mathsf{dim}\ W$, then no linear transformation from V to W is injective.

b. If $\mathsf{dim}\ V < \mathsf{dim}\ W$, then no linear transformation from V to W is surjective.

3. Prove that a function $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there exists a unique $m \times n$ matrix A such that $T(\alpha) = A\alpha$, for all $\alpha \in \mathbb{R}^n$. Let $L(\mathbb{R}^n, \mathbb{R}^m)$ denote the vector space of all linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$. Find a basis of $L(\mathbb{R}^n, \mathbb{R}^m)$.

4. Let T be a linear transformation from V to W. Suppose $\beta_1, \ldots, \beta_\ell \in V$ are linearly independent vectors that are not in $\mathsf{Ker}(T)$. Is it the case that $T(\beta_1), \ldots T(\beta_\ell)$ are also linearly independent? If not, give an explicit counter-example. If they are independent, prove it.

5. Let V be an n-dimensional vector space over the field F, and let $B = \{\alpha_1, \ldots, \alpha_n\}$ be an ordered basis of V. Define a linear operator on V as follows:

$$T(\alpha_j) = \alpha_{j+1}, \ j \in \{1, \ldots, n-1\},$$
$$T(\alpha_n) = 0.$$

Find the matrix representation of T in the ordered basis B. Also, prove that $T^n = 0$, but $T^{n-1} \neq 0$.

6. Let T be the linear operator on $\mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_2, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

What is the matrix representation of T in the standard ordered basis of $\mathbb{R}^3$? Also, compute the matrix of T in the ordered basis $\{(1, 0, 1), (-1, 2, 1), (2, 1, 1)\}$.

7. Prove the following statements about similarity of matrices.

a. Show that the identity matrix I is the only matrix that is similar to I.

b. Show that the zero matrix 0 is the only matrix that is similar to 0.

c. Suppose A and B are similar matrices. Prove that A is invertible if and only if B is invertible.

d. Show that similarity is an equivalence relation.

i. [Based on Example 11, Chapter 3, from Hoffman and Kunze] Let F be a subfield of complex numbers. Let V be the space of polynomials functions over F. Let D be the differentiation operator and let T be the "multiplication by x" operator, i.e. T maps a polynomial $p(x)$ to $xp(x)$.

a. Show that D is singular.

b. Find a right inverse for D.

c. Suppose f is a polynomial. Show that if $x \cdot f(x) = 0$ for all x, then $f(x) = 0$.

d. Show that T is non-singular.

e. Show that $DT - TD = I$.

ii. Solve Example 11 in Chapter 3 from the textbook (whatever was not covered in the question above).

iii. Let V be the space of infinite sequences

$$V := \{(x_1, x_2, x_3, \ldots) : x_1, x_2, x_3 \cdots \in \mathbb{R}\}.$$

Define the "left shift operator" $T : V \to V$ that maps $(x_1, x_2, x_3 \ldots)$ to $(x_2, x_3, \ldots)$.

i. Show that T is a linear operator, it is singular (i.e. has a non-trivial null-space) and yet, its image is V.

ii. How would your conclusion change, had you considered $T_k := TT \dots T$ (k-many times) for $k \in \{1, 2 \dots\}$, instead of T?

iii. How would you define a "right shift operator" S? Can you define it in such a way that S is a right inverse of T, i.e. $TS = \mathbb{I}$.

iv. Is S unique? If so, prove it. Otherwise construct distinct S and S' such that $TS = TS' = \mathbb{I}$.

v. Is $\mathsf{Im}(S) = V$? What about $\mathsf{nullspace}(S)$?

iv. Suppose T is a linear transformation from V to W where $\dim V = \dim W$. Given that $\alpha_1, \dots, \alpha_n$ form a basis for V, answer the following.

i. Disprove that $T\alpha_1, \dots, T\alpha_n$ is a basis for W (to disprove, give a T that is a counter-example)?

ii. Show that if T is onto, $T\alpha_1, \dots T\alpha_n$ forms a basis for W?

iii. Show that if T is non-singular, then again, $T\alpha_1, \dots T\alpha_n$ forms a basis.

v. Show that the following are equivalent, for a linear operator T.

i. Kernel of T contains only the zero vector.

ii. Nullity of T is zero.

iii. T is 1:1.

Does your answer work for infinite dimensional vector spaces as well?

vi. Let V be a vector space and suppose $\beta_1, \dots \beta_\ell \in V$ are linearly independent vectors not in the subspace $V' \subseteq V$. Is it the case that $\mathsf{span}(\beta_1, \dots \beta_\ell) \cap V = \vec{0}$? If so, prove it. Otherwise, write down the simplest counter-example you can conceive of.

vii. Suppose T is a linear operator on the vector space V. Let $\mathcal{B} := (\alpha_1, \dots \alpha_n)$ and $\mathcal{B}' := (\beta_1, \dots, \beta_n)$ are two bases for V. If $T\alpha_i = \beta_i$ for all i, show that T is invertible.